RESEARCH STATEMENT

EDWARD VARVAK

1. Summary of my results

My research concerns the structure of Lagrangian fibration on symplectic varieties.

Definition 1.1. A normal variety X is **symplectic** if it admits a non-degenerate symplectic form $\sigma \in H^0(X^{\mathrm{sm}}, \Omega^2_{X^{\mathrm{sm}}})$ which extends holomorphically onto a resolution of singularities $\pi : Y \to X$. In this setting, a **Lagrangian fibration** is a proper surjective morphism $f : X \to B$ with connected half-dimensional fibers, such that $\sigma|_{X_b} = 0$ for any fiber X_b.

We will need to assume that X is compact for most of our results. Additionally, we will want X to be "irreducible" in a sense, which will be made more precise in (*** link below). To that end, the most general class of symplectic varieties we will work with is the primitive symplectic varieties.

Definition 1.2. A compact symplectic variety X is **primitive symplectic** if it satisfies $H^1(X, \mathcal{O}_X) = 0$ and σ is unique up to scaling, i.e. $H^0(X, \Omega^2_X) = \mathbb{C}\sigma$.

As we will see (*** link below), when X is a primitive symplectic variety admitting a Lagrangian fibration $f : X \to B$, the smooth fibers X_b are Abelian varieties. My main results further classify the structure of the fibers.

Theorem 1.3. *Let X be a primitive symplectic variety of dimension $2n$ admitting a Lagrangian fibration $f : X \to B$. If f is not isotrivial, the Mumford–Tate group of the very general fiber X_b is isogenous to $(\mathrm{Sp}_{2h})^{n/h}$.*

Recall that an Abelian variety is Hodge-generic if it admits no extra Hodge tensors. In this case, it is simple and carries no extra symmetries.

Corollary 1.4. *The very general fiber X_b is isogenous to a product of Abelian varieties $A_1 \times \cdots \times A_{n/h}$, where each A_i is Hodge-generic and A_i is not isogenous to A_j for $i \neq j$.*

The most well-studied class of primitive symplectic varieties consists of hyper-Kähler manifolds, which should be thought of as the natural generalization of K3 surfaces.

Definition 1.5. A compact complex manifold X is **hyper-Kähler** if X is simply connected and its algebra of global holomorphic forms is generated by a non-degenerate symplectic form: $H^0(X, \Omega^{2p}_X) = \mathbb{C}\sigma^p, \quad H^0(X, \Omega^{2p+1}_X) = 0$.

Theorem 1.6. *Let $\mathcal{X}$ be a deformation class of hyper-Kähler manifolds such that*

1. *all members $X \in \mathcal{X}$ have second Betti number $b_2(X) \geq 7$,*
2. *there exists some $X \in \mathcal{X}$ admitting a Lagrangian fibration.*

Choose X to be general among the hyper-Kähler manifolds admitting a Lagrangian fibration, $f : X \to B$. If X_b is a very general fiber of f, then it has Mumford–Tate group $\mathbf{MT}(X_b) = \mathrm{GSp}_{2n}$, i.e. X_b is Hodge-generic.

These results largely came about through understanding the monodromy of the family f. To be more precise, let $f : X^\circ \to B^\circ$ be the restriction of f to its regular points, and let $V = R^1 f_* \mathbb{Z}_{X^\circ}$, viewed as a local system.

Theorem 1.7. *Suppose that X is primitive symplectic, $f : X \to B$ a Lagrangian fibration, $V_\mathbb{C} = V \otimes_\mathbb{Z} \mathbb{C}$ the underlying $\mathbb{C}$-representation of $\pi_1(B^\circ)$.*

(1) *If f is not isotrivial, then $V_\mathbb{C}$ is irreducible.*
(2) *If f is isotrivial, then its smooth fibers are isogenous to E^n, for E an elliptic curve. Furthermore, if K is the endomorphism field of E (possibly $\mathbb{Q}$), then $V_K = U_1 \oplus U_2$ splits into two geometrically irreducible components.*

Remark 1.8. There are a few clarifications to be made regarding the results and contributions. For the non-isotrivial case (1), the later result of Theorem 1.3 determines that a finite-index subgroup of the image of $\pi_1(B^\circ)$ lies in $(\mathrm{Sp}_{2h,\mathbb{Q}})^{n/h}$, and acts on $V_\mathbb{C}$ via a direct sum of standard representations. Such a representation is would not be irreducible, if not for the unaccounted finite part of the monodromy transitively permuting the irreducible factors. For the isotrivial case (ii), the result $X_b \sim E^n$ was first proven in (*** KLM), though we produced an alternative proof of this fact.

2. Background and motivation

Hyper-Kähler manifolds arise naturally in the study of K-trivial varieties. The following result is a classical theorem of Beauville–Bogomolov:

Theorem 2.1 (Beauville–Bogomolov Decomposition Theorem)**.** *Let Z be a compact complex manifold with $\omega_Z \simeq \mathcal{O}_Z$. Then there is a finite étale cover $\pi : Z' \to Z$ such that Z' has the form*

$$Z' = T \times \prod_i X_i \times \prod_j Y_j,$$

where T is a complex torus, the X_i are hyper-Kähler manifolds, and the Y_j are irreducible Calabi–Yau manifolds.

This theorem has since been generalized to allow for Z to have klt singularities (Bakker–Guenancia–Lehn); in this case, the X_i are irreducible holomorphic symplectic varieties, a condition stronger than primitive symplectic but weaker than hyper-Kähler.

Thus the problem of understanding families of K-trivial varieties boils down to understanding families of complex tori, hyper-Kählers, and ICYs. The former is largely classical, while the latter is also somewhat well-studied, thanks to work of Laza–Friedman (loc. cit.***) In contrast, we have a relatively poor understanding of the deformation classes of hyper-Kähler manifolds.

Given a polarized hyper-Kähler manifold (X, H), one can construct its deformation space as a type IV Shimura variety: take $L = H^\perp \subsetneq H^2(X, \mathbb{Z})$, define

$$\mathcal{D}_H = \{[\sigma] \in \mathbb{P}L_\mathbb{C} \mid \sigma \wedge \sigma = 0, \quad \sigma \wedge \overline{\sigma} > 0\}.$$

Then the deformation space of X is constructed as

$$\mathcal{X} = \Gamma \backslash \mathcal{D} = \Gamma \Big\backslash \tfrac{SO(2,b_2-3)}{SO(2)\times SO(b_2-3)},$$

where $b_2 = \dim H^2(X, \mathbb{C})$.

As such, to constuct a deformation space of hyper-Kähler manifold, it suffices to exhibit one member. There are currently four known deformation types: the infinite families $K3^{[n]}$ and Kum_n, occuring in every even dimension $2n$, and the two sporadic types $OG6$ and $OG10$ (loc. cit. O'Grady), occuring in dimensions 6 and 10 respectively.

Now consider the following central conjecture.

Conjecture 2.2. *(Hyper-Kähler SYZ Conjecture) Let $\mathcal{X}$ be a deformation space of hyper-Kähler manifolds. There exists some $X \in \mathcal{X}$ which admits a Lagrangian fibration $f : X \to B$.*

If true, one could make statements about all deformation classes of hyper-Kählers by simply proving them for those admitting Lagrangian fibrations. A recent paper of (engel 25***) assumed the SYZ conjecture to show that there are only finitely many deformation classes of hyper-Kählers in any fixed dimension.

As such, I believe it important to understand the structure of Lagrangian fibrations. Incidentally, it turns out that they carry many constraints.

Theorem 2.3. *Let X be a hyper-Kähler manifold of dimension $2n$, $f : X \to B$ a proper surjective morphism with connected fibers, $B \neq \{p\}$. The following all hold.*

[Greb Lehn 14] f is a Lagrangian fibration.
[Matsushita 99] The smooth fibers of f are Abelian varieties.
[Matsushita 14] B is $\mathbb{Q}$-Fano, $\mathbb{Q}$-factorial, and klt with Picard rank 1. There is an isomorphism $\mathrm{IH}^*(B, \mathbb{Q}) \simeq H^*(\mathbb{P}^n, \mathbb{Q})$.
[Hwang 08] If B is smooth, then $B \simeq \mathbb{P}^n$.
[Bakker] If f is not isotrivial, then it varies maximally, i.e. the period map $\mathcal{P} : B^\circ \to \mathcal{A}_n$ has an n-dimensional image.

My results Corollary 1.4, Theorem 1.6 aim to strenthen (2). Furthermore, as we will see in §3, this may yet have consequences for (3). The base B is a common area of study in this field, due to the following conjecture of Matsushita, partially resolved due to Hwang in (4).

Conjecture 2.4 (Matsushita)**.** *Let X be a hyper-Kähler manifold, $f : X \to B$ a Lagrangian fibration. Then $B \simeq \mathbb{P}^n$.*

3. Future directions

3.1. **A geometric characterization for the Mumford–Tate splitting.** Let $f : X \to B$ be a Lagrangian fibration, and suppose the Mumford–Tate group of the very general fiber X_b is of the form

$$\mathbf{MT}(X_b) \sim (\mathrm{Sp}_{2h})^k, \quad k \neq 1. \tag{3.1}$$

It is not hard to see that such a Lagrangian fibration exists. For example, we can take $g : S \to \mathbb{P}^1$ to be an elliptically fibered K3 surface; the Hilbert scheme of points $X := S^{[n]}$ is hyper-Kähler, and when g is not isotrivial, X admits a Lagrangian fibration

$$f = g^{[n]} : X \to \mathbb{P}^n, \quad \mathbf{MT}(X_b) \sim (\mathrm{SL}_2)^n.$$

More generally, if we allow X to be primitive symplectic, it is easy to see that we can attain every positive integer pair (h, k) in (3.1): by Theorem 1.6, we can find a Lagrangian fibration $g : Y \to \mathbb{P}^h$ with $\mathbf{MT}(Y_b) \sim \mathrm{Sp}_{2h}$, and taking the k-th symmetric power of this morphism produces

$$X = Y^{(k)} \xrightarrow{g^{(k)}} (\mathbb{P}^h)^{(k)},$$

a Lagrangian fibration realizing the Mumford–Tate group $(\mathrm{Sp}_{2h})^k$.

However, notice that the base $(\mathbb{P}^h)^{(k)}$ is easily distinguished from projective space when $h \neq 1$. This ensures that the argument above does not provide a *hyper-Kähler* manifold X realizing the prescribed Lagrangian fibration $f : X \to B$.

Question 3.1. *Does there exist a hyper-Kähler manifold (or even an irreducible holomorphic symplectic variety) X admitting a Lagrangian fibration $f : X \to B$ such that $\mathbf{MT}(X_b) \sim (\mathrm{Sp}_{2h})^k$ for some $h \neq 1, n$?*

I expect the answer to be negative. To motivate this somewhat, let $V = R^1 f_* \mathbb{Z}_{X^\circ}$ be the local system underlying f. I claim there exists a ramified cover $\pi : B' \to B$ which resolves the finite irreducible part of the monodromy, as referenced in Remark 1.8. To be more precise, $\pi^* V_{\mathbb{Q}} = U_1 \oplus \cdots \oplus U_k$, and the monodromy action on each U_i factors through the standard representation of $\mathrm{Sp}_{2h,\mathbb{Q}}$.

Now, using some methods from (bakker-schnell***), we can extend the isomorphism $\mathcal{T}_{B^\circ} \xrightarrow{\sim} F^1\mathcal{V}$ to the whole tangent bundle $\mathcal{T}_{B^{\mathrm{sm}}}$. By a local computation, we can pull this isomorphism back to get a splitting of the tangent bundle of B':

$$\mathcal{T}_{B'^{\mathrm{sm}}} \simeq F^1\overline{\mathcal{U}}_1 \oplus \cdots \oplus F^1\overline{\mathcal{U}}_k.$$

Another result of (bakker-schnell***) shows that each $F^1\overline{\mathcal{U}}_i$ is closed under the Lie bracket, thus giving a foliation of B'. By some more Hodge theory, we can actually show that the general leaf of this foliation is algebraic. Altogether, we can show that this produces the following diagram:

$$\begin{array}{ccc} X' & & X \\ \downarrow{\scriptstyle f'} & & \downarrow{\scriptstyle f} \\ B' & \xrightarrow{\pi} & B \\ \vdots{\scriptstyle \varphi^k} & & \vdots{\scriptstyle \varphi^{(k)}} \\ Z^k & \xrightarrow{\overline{\pi}} & Z^{(k)} \end{array}$$

Now we can rephrase Question 3.1 as follows:

Question 3.2. *Does the map $\varphi^{(k)} : B \dashrightarrow Z^{(k)}$ extend to a morphism?*

If indeed it does, then we must have an inclusion on the level of intersection cohomology:

$$\mathrm{IH}^*(Z^{(k)}, \mathbb{Q}) = \mathrm{Sym}^k(\mathrm{IH}(Z, \mathbb{Q})) \subseteq \mathrm{IH}^*(B, \mathbb{Q}) = H^*(\mathbb{P}^n, \mathbb{Q}).$$

But $H^*(\mathbb{P}^n, \mathbb{Q})$ is minimal among n-dimensional Kähler manifolds, so we must have equality everywhere. This is impossible unless $\dim Z = 1$, so that $\mathrm{IH}^*(Z, \mathbb{Q}) = H^*(\mathbb{P}^1, \mathbb{Q})$, which forces $Z \simeq \mathbb{P}^1$. This immediately gives $h = 1$ and $k = n$, thereby providing a negative answer to Question 3.1.

Another related question worth pursuing may also be seen in this diagram:

Question 3.3. *Is the map $\varphi^{(k)} : B \dashrightarrow Z^{(k)}$ birational?*

Suppose it were. By a theorem of Saccá (***), we can compactify the map $\varphi^{(k)} \circ f : X \dashrightarrow Z^{(k)}$ to a morphism $g : Y \to Z^{(k)}$, where Y is a smooth hyper-Kähler manifold birational to X, and $g : Y \to Z^{(k)}$ is a Lagrangian fibration agreeing with $\varphi^{(k)} \circ f$ away from the indeterminacy locus.

Now by a similar intersection cohomology argument, we have $\mathrm{IH}^*(Z,\mathbb{Q}) = \mathrm{IH}^*(\mathbb{P}^1,\mathbb{Q})$, forcing $h = 1$ and $k = n$, providing an affirmitive answer to Question 3.1. However, in this setting, we may be able to learn more about X. Over $\mathbb{P}^1$, we may be able to take the variation of Hodge structures $F^1\mathcal{U}_i$ and produce a K3 surface $S \to \mathbb{P}^1$ realizing the VHS. If so, we can show that $Y \simeq S^{[n]}$, providing an affirmitive answer to the following question:

Question 3.4. *Let X be a hyper-Kähler manifold, $f : X \to B$ a Lagrangian fibration. Suppose $\mathbf{MT}(X_b) \sim (\mathrm{Sp}_{2h})^k$, for $h < n$. Does X belong to the deformation space $K3^{[n]}$?*

It is possible that Question 3.3 has a negative answer. In this case, it may still be possible to attack Question 3.4 in other ways. For instance, we may be able to show that the tangent bundle $\mathcal{T}_X$ splits in the same way the local system does. We can try to integrate the resulting distributions, which could give us a handle on how to construct X using lower-dimensional hyper-Kähler manifolds.

3.2. **Towards the hyper-Kähler Bogomolov–Mumford conjecture.** Recall the classical theorem of Bogomolov–Mumford: an ample linear series on a K3 surface contains a divisor supported on rational curves. An analogue of this theorem remains open for hyper-Kähler manifolds.

Conjecture 3.5 (Hyper-Kähler Bogomolov–Mumford Conjecture)**.** *Let X be a hyper-Kähler manifold, L an ample line bundle on X. Then there exists a uniruled divisor $D \in |L|$.*

Currently, this conjecture is only known in full for the deformation types $K3^{[n]}$ and Kum_n. Currently, I do not even have an idea on how to attack it in full generality, but Theorem 1.6 may be useful in one aspect of this conjecture: producing an abundence of rational curves.

If we start with X a general hyper-Kähler admitting a Lagrangian fibration $f : X \to B$ as in the assumptions of Theorem 1.6, the image of the period map $B^\circ \dashrightarrow \mathcal{A}_n$ will be n-dimensional. On the other hand, the product locus $\mathcal{A}_1 \times \mathcal{A}_{n-1} \subseteq \mathcal{A}_n$ has codimension $n-1$. The period image is Hodge-generic, so by Ax–Schanuel, it must meet $\mathcal{A}_1 \times \mathcal{A}_{n-1}$ in the expected dimension $n - (n-1) = 1$.

In other words, there is a curve $C^\circ \subseteq B^\circ$ over which the family of Abelian varieties $X_{C^\circ} \to C^\circ$ is isogenous to a product family. It should be reasonable to show that over the boundary of C°, the elliptic part of the family degenerates into a rational curve, while the non-elliptic part remains non-degenerate. This gives rise to a uniruled algebraic cycle of dimension $n+1$ in $\mathrm{CH}^*(X)$.

Thus, we have methods of producing rational curves in the singular fibers of f. However, we want to produce rational curves on all hyper-Kähler manifolds, not simply the ones admitting Lagrangian fibrations. If we assume Conjecture 2.2, this reduces to the following question:

Question 3.6. *Let X be a hyper-Kähler manifold admitting a Lagrangian fibration $f : X \to B$. Let $Z \subseteq X$ be a singular rational curve contained in a singular fiber of f, obtained as above. Let X' be a general deformation of X; does Z deform to a rational curve $Z' \subseteq X'$?*

Finally, if we truly hope at attacking Conjecture 3.5, we would want to break a divisor $D \in |L|$ into the union $D \sim D^{\mathrm{vert}} + D^{\mathrm{base}}$, where D is a uniruled divisor which is vertical

with respect to the Lagrangian fibration, and D^{base} is a uniruled divisor which is pulled back from the base.

Department of Mathematics
University of Chicago
5734 S University Ave
Chicago, IL 60637

Email address: varvak@uchicago.edu